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AI Protectionism, Energy and Semiconductors:
US/Europe Divergence Trajectories 2024–2030

Integrated Geostrategic and Economic Analysis

Chapter V

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77% global AI compute = USA **1.6×** energy cost EU/US **3.4:1**
CACI ratio US/EU (Power Mode)

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7 chapters • 4 prospective scenarios • 3 geographic zones

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CHAPTER V

Prospective Scenarios 2026–2030

This chapter constitutes the core of this study's original contribution. By applying the methodological protocol defined in Chapter II (2×2 matrix, six divergence metrics, CACI calibration), we construct four scenarios for the evolution of the transatlantic relationship in AI, energy, and semiconductors for the 2026–2030 period. Each scenario is determined by the combination of two critical uncertainties identified in Chapter III: the degree of American protectionism and the capacity for European strategic response. We then evaluate each scenario on its six metrics, before synthesizing the tipping conditions between trajectories.

5.1 Predetermined Elements: What Will Not Change

In accordance with the Schwartz method (1991), we distinguish predetermined elements (near-certain trends by the 2030 horizon) from critical uncertainties (factors whose evolution depends on political decisions not yet made). Four predetermined elements structure all scenarios.

PE1 — Exponential growth of AI compute demand. Semiconductor sales doubled in two years (2023–2025), installed AI chip power doubles every seven months (Epoch AI), and no sign of slowdown is observable as of February 2026. Even under the hypothesis of scaling law deceleration ("Chinchilla saturation"), the diffusion of AI toward inference, robotics, and autonomous agents will maintain strongly rising compute demand.[1]

PE2 — Persistent concentration of compute in the United States. The US/EU ratio of ~16:1 in raw installed compute (Chapter III, Epoch AI data: 2,763,554 vs 173,416 PF), translating to a CACI Power Mode ratio of 3.4:1 (weighted geometric formula), reflects investment decisions made in 2022–2025, whose effects materialize through 2028–2029 (data center construction timelines: 18–36 months). Even an immediate policy reversal would not alter the installed stock before the end of the decade.

PE3 — Growing energy tension. Global data center consumption, estimated at 415 TWh in 2024, will reach 800–950 TWh by 2030 according to IEA projections (Chapter III). The asymmetry in energy costs (US 2–3× cheaper than the EU) will persist, barring massive investment in European nuclear power — whose deployment timelines (SMR: 5–7 years for initial reactors) extend beyond the 2030 horizon.[2]

PE4 — Section 232 regulatory framework in place. Proclamation 11002 of January 14, 2026 is a legal fait accompli. Unlike IEEPA tariffs (struck down by the Supreme Court on February 20, 2026[3]), Section 232 tariffs rest on a confirmed legal basis. The Secretary of Commerce's report on the semiconductor market for data centers is expected by July 1, 2026, and may recommend an expansion or modification of tariffs. Regardless of the direction taken, the legal instrument will remain available.[4]

5.2 Critical Uncertainties and the 2×2 Matrix

5.2.1 Axis 1: Degree of American Protectionism

The first uncertainty concerns the evolution of American policy between two poles. The "moderate" pole corresponds to maintaining the January 2026 status quo: 25% tariff limited to re-exported advanced chips, broad domestic exemptions, EU trade agreement capping semiconductor tariffs at 15%, and no significant extension to cloud or models. The EU "core" (France, Germany) remains in the trusted partner category. The "aggressive" pole assumes an expansion after the July 2026 report: tariffs extended to derivative semiconductors and equipment, GPU quotas for the EU (including France), restrictive conditions for access to cutting-edge AI cloud, and use of compute as a trade negotiation lever ("compute-for-concessions").[5]

5.2.2 Axis 2: European Strategic Response Capacity

The second uncertainty concerns the EU's ability to deploy a coherent and rapid response. The "reactive" pole corresponds to fragmented national responses, dispersed investments, an AI Act creating additional compliance costs, and slow deployment of AI Factories/Gigafactories (bureaucratic delays, 24+ month permitting). The "proactive" pole assumes accelerated implementation of the AI Continent program (19 AI Factories + up to 5 Gigafactories of 100,000+ GPUs), adoption of Special Compute Zones (180-day permitting), effective mobilization of the InvestAI fund (€20 billion), and pooling of French nuclear capacity as a competitive advantage.[6]

5.2.3 Matrix and Scenario Naming

The intersection of these two axes produces four scenarios:

	Reactive EU Response	Proactive EU Response
Moderate US Protectionism	**Scenario A** "Reinforced Status Quo" Slow drift toward dependency	**Scenario C** "Asymmetric Partnership" Western tech junior partner
Aggressive US Protectionism	**Scenario B** "Digital Fracture" Structural European decoupling	**Scenario D** "Contested Sovereignty" Autonomy race under pressure

Table 10. 2×2 matrix of prospective scenarios 2026–2030. Source: author's construction, Schwartz (1991) methodology.

5.3 Scenario A — "Reinforced Status Quo" (Moderate Protectionism + Reactive EU)

5.3.1 Narrative

After the July 2026 report, the Secretary of Commerce recommends maintaining the 25% tariff on re-exported advanced chips but not extending it significantly. The August 2025 US-EU trade agreement is respected: semiconductor tariffs for the EU remain capped at 15%.[7] The EU, reassured by this status quo, slows the deployment of its own initiatives. EuroHPC AI Factories struggle to reach their nominal capacity (permitting delays, inter-state coordination). Gigafactories are pushed back to 2029–2030. The InvestAI fund is partially mobilized (€8–10 billion out of 20). European companies continue to rely heavily on US cloud, whose performance and costs remain unbeatable.

5.3.2 Metrics Trajectory

M1 — Compute ratio (installed GPUs US/EU): increases from 3.4:1 (2025) to 4–5:1 (2030). The gap widens slightly as US investments accelerate (Stargate, xAI mega-clusters, Meta) while the EU adds only the 19 AI Factories (25,000 GPUs max each, i.e., ~475,000 public GPUs, an order of magnitude below a single US hyperscaler).[8]

M2 — FLOP cost gap (EU/US): remains in the 2.4–3.2× range. The absence of aggressive tariffs on the EU maintains access to US cloud at prices close to current levels, but European energy costs continue to weigh.

M3 — US cloud share in European AI spending: increases from 70% (2024) to 72–75% (2030). European providers (OVHcloud, Deutsche Telekom) maintain their 15% share on the sovereignty segment but do not gain ground on generative AI services.

M4 — AI productivity (%/year): US: +2.5–3.0%; EU: +1.0–1.5%. The EU realizes part of the AI potential through downstream applications (SAP, Siemens, fintech), but slow adoption and the compute deficit cap gains.

M5 — Energy dependency (TWh data centers): EU: ~115 TWh (2030, i.e., +65% vs 2024). French nuclear energy absorbs part of the demand, but the absence of Special Compute Zones delays the connection of new data centers to the grid.

M6 — CACI(US)/CACI(EU): increases from 3.4:1 (2025) to 4–5:1 (2030). The gap widens moderately as the F(r) (compute) factor increases on the US side while energy costs E(r) weigh on the EU side.

5.3.3 Consequences for France

This scenario is the most likely in the short term (estimated probability: 40–50%). It is also the most insidious: the absence of a visible shock demobilizes European actors, while dependency deepens structurally. French companies benefit from access to US cloud for AI adoption (BNP Paribas, Airbus, TotalEnergies via AWS/Azure), but this adoption reinforces the lock-in described in Chapter IV. The AI productivity deficit relative to the United States (−1.0 to −1.5 points per year) accumulates over five years, widening the competitiveness gap by 5 to 8 GDP points.

5.4 Scenario B — “Digital Fracture” (Aggressive Protectionism + Reactive EU)

5.4.1 Narrative

The July 2026 report leads to a significant expansion. The Secretary of Commerce recommends tariffs extended to semiconductor equipment and derivative products, with a tariff offset program reserved for companies investing in American production.[9] The EU’s 15% agreement is revised upward, or accompanied by restrictive conditions (volume quotas on advanced GPUs, reciprocity requirements on the AI Act). Simultaneously, access to cutting-edge AI cloud is made conditional for non-American entities (limitations on API access to frontier models, restrictions on weights). The EU, fragmented, fails to formulate a coherent response: member states split between accommodation (Nordic countries, Netherlands) and confrontation (France, Italy).

5.4.2 Metrics Trajectory

M1 — Compute ratio: increases from 3.4:1 to 6–7:1 (2030). GPU quotas limit European imports at the moment when demand explodes. AI Factory projects are compromised by the inability to procure Nvidia/AMD GPUs at planned volumes.

M2 — FLOP cost gap: jumps to 4–6×. Expanded tariffs, combined with quotas and energy asymmetry, massively increase European compute costs. French companies face a 3× to 5× surcharge for model training.

M3 — US cloud share: paradoxically, rises to 78–82%. Lacking a credible local alternative, European companies wanting access to cutting-edge AI must go through US hyperscalers, at whatever pricing terms they dictate. Sovereign services (OVHcloud, Scaleway) lack the hardware to offer competitive GenAI services.

M4 — AI productivity: US: +2.5–3.5%; EU: +0.3–0.8%. European AI potential is severely constrained. The McKinsey Global Institute estimates that with slow adoption, European productivity would not exceed 0.3% — close to stagnation.[10]

M5 — Energy dependency: EU: ~95 TWh only (2030), not by virtue but by default — the lack of GPUs limits data center construction. Ironically, the compute constraint attenuates the energy constraint.

M6 — CACI ratio: explodes to 5–8:1 (2030). This is the scenario where the gap is largest, with all three CACI factors deteriorating simultaneously on the European side: F(r) capped by quotas, E(r) inflated by tariffs, L(r) weakened by accelerated brain drain to the United States.

5.4.3 Consequences for France

This scenario (estimated probability: 15–20%) represents the worst case. France suffers a structural technological decoupling: compute-intensive projects (Mistral foundation models, Comau/Exotec robotics, Dassault simulations) are relocated to the United States or dependent on increasingly costly US cloud access. The time-to-market of French AI solutions extends by 25 to 40%. Industrial SMEs, unable to absorb the surcharges, forgo cutting-edge AI and opt for degraded solutions (smaller open-source models, local inference). The cumulative productivity gap with the United States reaches 10 to 15 points over five years.

5.5 Scenario C — “Asymmetric Partnership” (Moderate Protectionism + Proactive EU)

5.5.1 Narrative

US protectionism remains moderate (as in A), but the EU exploits this window to accelerate its own investments. AI Factories are deployed on schedule (2026–2027), the first Gigafactories of 100,000+ GPUs are ordered in late 2026 and delivered in 2028.[11] France plays a central role thanks to its nuclear fleet (65–70% of the electricity mix, competitive marginal cost), and Special Compute Zones are designated on former industrial sites with heavy grid connections.[12] However, the EU accepts de facto a junior technological partner status: it uses Nvidia/AMD GPUs (no European champion in AI ASIC design), depends on TSMC/Samsung/Intel foundries for production, and its foundation models remain one step below US leaders.

5.5.2 Metrics Trajectory

M1 — Compute ratio: decreases from 3.4:1 (2025) to 2–2.5:1 (2030). Gigafactories and private investment (InvestAI + industrial co-investments) add 1–2 million GPU equivalents in Europe, reducing the gap without closing it.

M2 — FLOP cost gap: decreases to 1.5–2.0×. French nuclear and Gigafactory economies of scale compress energy and infrastructure costs, although a residual gap persists (absence of proprietary GPU design).

M3 — US cloud share: decreases slightly to 60–65%. European sovereign services gain market share on regulated segments (defense, healthcare, finance), while US cloud retains the majority of commercial workloads. The market segments into “sovereign” and “performance.”

M4 — AI productivity: US: +2.5–3.0%; EU: +1.8–2.5%. The EU reaches 60–80% of theoretical potential thanks to sufficient local compute for large-scale adoption of downstream applications, even though frontier model training remains dependent on US hardware.

M5 — Energy: EU: ~140 TWh (2030). Demand is higher than in A because European compute increases, but nuclear and planned SMRs absorb the bulk. RTE France confirms the feasibility of +10 GW subject to grid investments.

M6 — CACI ratio: decreases to 4–7:1 (2030). This is the most favorable scenario realistically achievable by the 2030 horizon. The C(r) factor improves significantly, E(r) benefits from nuclear, but L(r) remains slightly lower (US AI ecosystem more attractive for top talent).

5.5.3 Consequences for France

This scenario (estimated probability: 15–20%) is the most favorable for France in the short-to-medium term. France becomes the EU’s AI energy hub thanks to its nuclear fleet, attracting data center and Gigafactory investments. French companies gain access to competitive local compute for inference and fine-tuning, reducing dependency on US cloud for standard use cases. Mistral and French startups can train specialized models locally. However, frontier model training remains dependent on US hardware, and strategic autonomy is partial: France is sovereign in application, but not in the creation of foundational technologies.

5.6 Scenario D — “Contested Sovereignty” (Aggressive Protectionism + Proactive EU)

5.6.1 Narrative

US protectionism intensifies (as in B), but the EU responds with determination. The American threat becomes the political catalyst for an unprecedented European industrial mobilization since the AIRBUS project of the 1970s. The AI Continent program is accelerated and expanded: the 5 Gigafactories are urgently ordered, France announces 20 GW of nuclear capacity dedicated to AI data centers by 2032 (combining extension of the existing fleet and SMRs), the DARE project (European RISC-V) is escalated to design AI accelerators reducing dependency on Nvidia.[13] Simultaneously, the EU negotiates alternative technology alliances (Japan, South Korea, Taiwan) to secure GPU and foundry supply.

5.6.2 Metrics Trajectory

M1 — Compute ratio: evolves from 3.4:1 (2025) to 3–4:1 (2030). The EU invests massively but starts from very far behind. US quotas slow imports, but alternative alliances and local production (Gigafactories using Samsung/Intel GPUs as Nvidia alternatives) partially compensate.

M2 — FLOP cost gap: 2.5–4.0× initially (2027, peak of the tariff shock), then progressive reduction toward 1.8–2.5× (2030) as Gigafactories ramp up and GPU alternatives mature.

M3 — US cloud share: decreases to 50–55% (2030), the most pronounced decline of the four scenarios. Geopolitical distrust and US restrictions push European companies toward sovereign alternatives, even imperfect ones. US hyperscalers lose ground on regulated segments.

M4 — AI productivity: US: +2.5–3.5%; EU: +1.2–2.0%. The EU experiences a productivity trough in 2027–2028 (transition period when US restrictions bite but European investments are not yet operational), then a partial catch-up from 2029 onward.

M5 — Energy: EU: ~150–160 TWh (2030). This is the most energy-intensive scenario for the EU, as massive local data center construction creates enormous demand. French nuclear becomes a continental strategic asset, but grid pressure is at its maximum.

M6 — CACI ratio: follows a U-shaped trajectory: degradation to 15–20:1 in 2027–2028 (shock), then improvement toward 8–12:1 by 2030. The outcome depends heavily on European execution speed: every year of delay in Gigafactories extends the period of maximum vulnerability.

5.6.3 Consequences for France

This scenario (estimated probability: 15–20%) is the most ambitious and the riskiest. It places France at the heart of an unprecedented European technological sovereignty effort. Massive nuclear investments (SMR, fleet extension) become a first-order geopolitical issue. The DARE/RISC-V project could, if successful, constitute the first credible European alternative to Nvidia GPUs for AI — but on a 5–7 year horizon, well beyond 2030. In the short term (2026–2028), France traverses a period of maximum vulnerability where surcharges and shortages degrade competitiveness, before a catch-up conditional on infrastructure deployment speed.

5.7 Comparative Synthesis and Tipping Conditions

5.7.1 Summary Metrics Table

Metric (2030)	A — Status Quo	B — Fracture	C — Partnership	D — Sovereignty
M1 Compute ratio US/EU	4–5:1	6–7:1	8–10:1	8–12:1
M2 FLOP cost gap	2.4–3.2×	4–6×	1.5–2.0×	1.8–2.5×

M3 US cloud share (%)	72–75	78–82	60–65	50–55
M4 EU productivity (%/yr)	+1.0–1.5	+0.3–0.8	+1.8–2.5	+1.2–2.0
M5 EU energy (TWh)	~115	~95	~140	~155
M6 CACI ratio	3–4:1	5–8:1	4–7:1	8–12:1
Estimated probability	40–50%	15–20%	15–20%	15–20%

Table 11. Comparative summary of the four scenarios on the six divergence metrics. Source: author's construction.

[Graph G5]
CACI(US)/CACI(EU) Trajectories 2025–2030 by Scenario (4 curves)

5.7.2 Tipping Conditions Between Scenarios

The actual trajectory will likely follow a hybrid path between these scenarios. Three tipping points determine the possible transitions.

First tipping point: the Commerce report of July 2026. This report will determine whether US protectionism expands (tipping toward B or D) or remains targeted (remaining in A or C). Indicators to watch include: the evolution of the American semiconductor trade deficit, the fill rate of CHIPS Act fabs (Intel, TSMC Arizona, Samsung Taylor), and domestic political pressure (2026 midterms). The outcome of Phase 1 negotiations (report due April 14, 2026) will be an early signal.[14]

Second tipping point: the speed of EU Gigafactory deployment. The Commission plans the first operational Gigafactories in 2027–2028. If this timeline is met, the EU tips toward proactive scenarios (C or D). If permitting, financing, or hardware procurement delays push deliveries to 2029–2030, the EU remains in reactive mode (A or B). The CFG proposal for Special Compute Zones (180-day permitting vs. current 24+ months) is the key accelerating factor.[15]

Third tipping point: the French decision on nuclear for AI. France possesses a unique asset in Europe: a nuclear fleet providing 65–70% of electricity, with globally competitive marginal cost. The decision to dedicate significant capacity (10–20 GW) to AI data centers, through fleet extension, new EPR2s and SMRs, will determine whether France becomes Europe's AI energy hub or cedes this position to others (Scandinavia with hydroelectric power, Eastern Europe with low land costs). This tipping point is distinctly French and determines France's position within the European scenarios.[16]

5.7.3 The Convergence Point: 2028

The four scenarios converge on a common critical point in 2028. This is the year when: (i) compute demand will exceed installed capacity in Europe, creating material bottlenecks (even under moderate protectionism); (ii) the initial effects of expanded tariffs (if adopted) will be fully felt; (iii) Gigafactories, if deployed on time, will begin producing significant local compute; (iv) data center energy demand will saturate grid connection capacity in several member states. The year 2028 therefore constitutes the moment of truth when Europe will discover whether it is on trajectory A/B (growing dependency) or C/D (catch-up begun). The decisions made in 2026–2027 — US Commerce report, Gigafactories, French nuclear — will be irreversibly committed.

[Graph G6]
Tipping Points Chronology 2026–2030 and Decision Windows

5.8 Origins of the Tipping Point: Juridico-Technical Foundations Already in Place

The 'Great Decoupling' scenario is not built from thin air. It is the logical projection, to the 2028 horizon, of a control architecture whose foundational layers are already operational in 2026. Identifying these layers is a matter of academic rigor: distinguishing what falls under documented trends from what constitutes extrapolated foresight.

5.8.1 The Legal Layer: Extraterritoriality as a Structural Instrument

The first layer is legal and long predates AI policy. The CLOUD Act (Clarifying Lawful Overseas Use of Data Act, 2018) establishes that any cloud provider subject to US jurisdiction is required to produce data "regardless of where such communications, records, or other information are stored."[17] This statute, confirmed by federal case law (United States v. Microsoft, resolved by the CLOUD Act's enactment before the Supreme Court ruled), creates a fundamental dissociation between the physical location of data and its legal nationality.

The immediate consequence for AI compute is radical: an H100 cluster physically located in Dubai, Singapore, or an AWS eu-west-1 data center in Ireland remains legally American. In the event of a US government order, the operator — AWS, Azure, Google Cloud — is legally required to comply, regardless of the client's wishes or local legislation. Microsoft acknowledged before a French court in 2024 that it could not guarantee data sovereignty for European customers in the event of a legally founded US injunction.[18]

This pre-existing legal architecture is the substrate onto which the 'Cloud-Nationality' Pivot grafts itself: the Cloud Sovereignty Mandates projected for 2028 do not create a new power — they activate and systematize an already-existing jurisdictional power by extending it to the operational compute layer.

5.8.2 The Technical Layer: From Location Verification to Cluster Throttling

The second layer is technical. The US AI Action Plan of July 23, 2025, explicitly introduces the concept of location verification features applied to advanced AI chips. Michael Kratsios, Director of the White House Office of Science and Technology Policy, confirmed: "There is discussion about potentially the types of software or physical changes you could make to the chips themselves to do better location-tracking. That is something we explicitly included in the plan."[19]

This announcement is not rhetorical. BIS already holds, since the *Framework for Artificial Intelligence Diffusion* (January 2025, rescinded in May then replaced by guidance measures), a compute-tiering control architecture for destination countries (Tier 1–2–3) and compute caps per entity and per country.[20] The BIS Affiliates Rule, suspended for one year in November 2025 but maintained in principle, stipulates that a company's affiliation with a parent entity in a restricted country is sufficient to deny it access to advanced compute — regardless of the physical location of the cluster.[21]

The technological trajectory toward 2028 is therefore: (i) chips equipped with location verification mechanisms (software or hardware), (ii) automatic reporting systems to BIS in case of deviation, (iii) capacity to suspend access or throttle performance via export license — a cluster operating under a US export license potentially subject to operational restrictions by administrative decision. This is not science fiction: it is the extension to compute of a principle already applied to software (OFAC sanctions on software licenses, freezing of cloud service access for listed entities).

Critical limitation of this mechanism. *A genuine technical tension must be identified here: throttling production clusters is technically complex and potentially disruptive for the operators themselves. NVIDIA does not currently have a remote-disabling mechanism for H100/B200 GPUs in production. Such a mechanism would require significant architectural modification to firmware and remote attestation protocols (via TPM or equivalent). The 2028 scenario is therefore conditional on the effective implementation of these modifications — a plausible hypothesis over 24–36 months of industrial effort, but not a certainty.*

5.9 The Mechanism of the "Cloud-Nationality" Pivot

5.9.1 The Trigger: Cloud Sovereignty Mandates as an Extension of Export Controls

The 2028 tipping-point scenario assumes the United States takes a qualitative step forward: moving from hardware access control (BIS export controls on chips) to operational compute service access control (Cloud Sovereignty Mandates). This step is not an arbitrary rupture — it responds to a structural flaw identified in the current control regime.

This flaw is documented: despite BIS restrictions on H100/A100 exports, investigations revealed that approximately $1 billion worth of Nvidia chips were routed to China circumventing export controls via third countries (Malaysia, UAE, Singapore) in the first months of 2025 alone.[22] The AI Action Plan's response — location verification features and cluster monitoring — constitutes the first step toward continuous post-export control.

The plausible trigger in 2028 is an executive order extending the obligations of the *Framework for AI Diffusion* to the cloud layer. Its structure would be as follows:

- Any commercial operation of AI clusters using H100/B200/equivalent Blackwell chips (ECCN 3A090.a) by a US hyperscaler on foreign soil is subject to a 'Data Residency & Jurisdiction Compliance' (DRJC) program;
- US hyperscalers (AWS, Azure, GCP) are required to certify that their offshore clusters are not operating for the benefit of entities in Tier 3 countries or under unauthorized Tier 2 entity control — on pain of revocation of their access to advanced NVIDIA/AMD compute on US soil;
- BIS reserves the right to reduce authorized cluster performance (compute throttling via license) if compliance irregularities are detected, up to full suspension.

5.9.2 The Dissociation of Physical Factor / Sovereign Factor in the CACI

This is where the scenario produces its most analytically significant impact on the CACI model developed in Chapters III and IV. The current model integrates compute C(r) as a measure of physically installed capacity in region r. Under activation of the Cloud-Nationality Pivot, however, the variable C(r) splits into two distinct components:

C(r) = C_phys(r) × F_sov(r)

C_phys(r) = compute physically installed in jurisdiction r (H100-equivalent GPUs, TWh consumed) — the measure used in Chapters III–V;

F_sov(r) = operational sovereignty factor, defined as the fraction of compute in C_phys(r) that is *outside US jurisdiction* and therefore insensitive to Cloud Sovereignty Mandates ($0 \leq F\_sov \leq 1$);
F_sov = 1 − F_US(r) where F_US(r) is the fraction of regional compute operated by hyperscalers subject to the CLOUD Act.
Calibrated estimates for 2028, under the hypothesis of Cloud Sovereignty Mandate activation, are as follows:

Jurisdiction	C_phys (US cloud share)	F_sov estimate	Current CACI (Physical Factor)	Corrected CACI (Sovereign Factor)
United States	~5% (domestic cloud — not affected)	**1.00**	CACI(US) = reference 100	**100 (unchanged)**
EU (France, Germany)	~77% (AWS/Azure/GCP dominate EU cloud)	**0.28**	CACI(EU) ≈ 29 (US/EU ratio = 3.4:1, Power Mode)	**30–50% score collapse**
UAE (Dubai hub)	~88% (AWS/Azure/GCP dominant)	**0.12**	CACI(UAE) high — regional physical hub	**60–80% collapse — illusory hub**
Singapore	~82% (dominant US hyperscalers)	**0.18**	CACI(SGP) high — Asia-Pacific hub	**55–75% collapse**
China	~2% (Alibaba/Tencent/Huawei Cloud)	**0.98**	CACI(CHN) low — advanced chip shortage	**Unchanged — sovereignty already effective**
India	~60% (AWS/Azure + local players)	**0.40**	CACI(IND) medium-low	**Moderate collapse — intermediate position**

Table 12. Estimated F_sov Sovereignty Factor by jurisdiction and CACI impact under Cloud Sovereignty Mandate activation (2028). Source: author's construction; cloud market share data: Synergy Research Group (2025), Statista Enterprise Cloud (2025).
This table reveals the central paradox of the scenario: the jurisdictions that have invested most heavily in attracting US hyperscaler data centers — UAE, Singapore, certain European hubs — are precisely those whose CACI score collapses most severely. Their high physical compute reflected a real capacity, but one held on lease:

under Mandate activation, that compute becomes conditional, potentially throttled or suspended by US administrative decision.

5.10 The Emergence of Jurisdictional AI Blocs (2028–2030)

The Great Decoupling does not produce a binary US/non-US world. It produces fragmentation into blocs of varying intensity, depending on each zone's capacity to develop credible sovereign compute. Four blocs emerge with distinct characteristics.

5.10.1 The Extended American Bloc (American AI Alliance)

The July 23, 2025 AI Action Plan explicitly lays the groundwork for an "American AI Alliance": export of the full US technology stack (hardware, models, software, standards) to willing allies, in exchange for adoption of aligned export controls.[23] The strategy is explicitly described as "carrot and stick": aligned allies access advanced chips and frontier models without additional restrictions; those who refuse are exposed to Foreign Direct Product Rule mechanisms and secondary tariffs.

Members of the Extended American Bloc (confirmed Tier 1): United States, United Kingdom, Canada, Australia, Japan, South Korea, Netherlands, Germany, France (subject to alignment on export controls). For these countries, effective F_sov increases: their entities access US hyperscalers without restriction, and sovereign compute under development (EU Gigafactories for Europe) receives preferential treatment. The CACI of these countries is not degraded by the Mandates — it may even benefit from an alliance effect.

5.10.2 The Eurasian Sovereign Bloc

China constitutes the only complete example of a pre-existing sovereign bloc. With an F_sov of ~0.98 and a national cloud ecosystem (Alibaba Cloud, Tencent Cloud, Huawei Cloud) operating outside US jurisdiction, the Cloud Sovereignty Mandates have no direct traction. China's constraint remains the shortage of advanced chips (2022–2025 export controls have limited access to H100/A100/B200 GPUs), but the American Bloc cannot throttle a Huawei Ascend 910B cluster.

The post-2028 dynamic: China holds the *only* large-scale sovereign compute outside the American bloc. Countries seeking to emancipate themselves from Cloud Sovereignty Mandates find themselves structurally facing a binary alternative: conditional American compute or Chinese compute under other forms of dependency. This binary constraint is the most profound geopolitical impact of the Great Decoupling.

5.10.3 The Digital Non-Aligned: An Untenable Position

India, Brazil, Southeast Asia, and Gulf countries (absent special treaties with the United States) constitute a bloc of digital non-alignment. Their position is structurally uncomfortable: too dependent on US hyperscalers to shift toward sovereignty, insufficiently integrated into the American alliance to escape restrictions in the event of geopolitical disagreement.

The UAE case illustrates this fragility. Dubai has invested heavily since 2022 to become a regional AI hub, notably through agreements with AWS, Microsoft, and G42. Yet G42 was already subjected to intense US pressure in 2024 to sever its ties with Chinese entities — a condition imposed by Washington for access to advanced chips.[24] Under Cloud Sovereignty Mandates, this pressure would become systemic: the compute of Gulf hubs, physically present but legally American, would become a permanent negotiating lever.

5.10.4 The European Bloc: Between Alliance and Autonomy

Europe occupies an intermediate and evolving position. Legally Tier 1 (France, Germany, Netherlands, etc. are explicitly in the BIS presumption of approval for advanced chips), the EU nevertheless maintains a strategic autonomy ambition that the American Alliance does not fully satisfy.

The Cloud and AI Development Act (CADA), whose formal proposal is expected in Q1 2026, attempts to address this dilemma by defining an *EU Sovereignty Level* that would structurally exclude CLOUD Act-subject providers from sensitive public markets.[25] The European Commission published in October 2025 a *Cloud Sovereignty Framework* defining three assurance levels (SOV-1 to SOV-3), with SOV-3 requiring that the provider be beyond the reach of any non-European extraterritorial law.[26]

This legislative architecture is under construction, but its timeline is problematic: CADA will at best be operational in 2027–2028, precisely when US Cloud Sovereignty Mandates could be activated. The vulnerability window is maximal between 2028 and 2030.

5.11 Cross-cutting Impacts on Scenarios A–D

The Cloud-Nationality Pivot overlays the four scenarios, modifying their conclusions in a non-linear fashion. It does not invalidate the 2×2 matrix but adds a third dimension: the degree of autonomy of the installed compute.

Scenario	Without Mandates	With 2028 Mandates	Impact on EU CACI	Strategic reading
A — Status Quo	Slow dependency, CACI ratio 4–5:1	Partial activation — hyperscalers cooperate without major restriction	Moderate 15–25% degradation — Tier 1 alliance partially protects	Most stable scenario — but security illusion revealed
B — Fracture	Structural decoupling, CACI ratio 5–8:1	Maximum activation — conditional US cloud simultaneously with restricted chips	**Double scissor effect: rare chips + conditional compute. CACI ratio potentially > 8:1**	Absolute worst case — chips + cloud sovereignty combo
C — Asymmetric Partnership	Partial catch-up, CACI 4–7:1	Sovereign Gigafactories absorb the shock if EU F_sov rises to ~0.45–0.55 via investments	Limited impact if deployed on schedule — Gigafactories = sovereign hedging	Best resilience — prior investment proves its worth
D — Contested Sovereignty	Catch-up under pressure, CACI 8–12:1	Mandates become the political catalyst — accelerate EU deployment and alternative alliances (JP, KR, TW)	Accelerated U-curve — 2028–2029 trough, then faster-than-forecast catch-up	Paradoxical: Mandates can accelerate EU sovereignty if response is fast enough

Table 13. Impacts of the Cloud-Nationality Pivot (Cloud Sovereignty Mandates 2028) on the four 2×2 matrix scenarios. Source: author's construction.

5.12 Implications for France: The Question of Truly Sovereign Compute

The Great Decoupling reframes a question that Scenarios A–D addressed implicitly: when France and Europe speak of 'sovereign compute,' what exactly do they mean? The Physical Factor / Sovereign Factor distinction reveals that current policy conflates two distinct objectives.

5.12.1 'Sovereignty Washing' as a Systemic Risk

The term *sovereignty washing* — popularized by Cristina Caffarra (Eurostack Foundation) — refers to the practice of US hyperscalers marketing 'sovereign cloud' offerings by implanting data centers on European soil, while remaining subject to the CLOUD Act.[27] Microsoft acknowledged in its own commercial documentation that it cannot guarantee sovereignty for European customers in the event of a legally founded injunction in the United States.[28]

The Cloud Sovereignty Framework published by the Commission in October 2025 begins to formalize this distinction — the SOV-3 level explicitly excludes entities subject to non-European extraterritorial legislation. But if maintained in the final CADA text, this requirement would de facto exclude AWS, Azure, and GCP from the most sensitive public markets — a politically conflictual decision vis-à-vis the United States.

5.12.2 France's Nuclear Advantage in the New Equation

The sovereigntist reinterpretation of the CACI paradoxically reinforces France's strategic asset. If C(r) decomposes into C_phys × F_sov, then France's optimal strategy is not only to increase C_phys (attract more hyperscaler data centers) but to increase F_sov (develop compute independent of US jurisdictions).

French nuclear energy creates a first-order competitive advantage here: Special Compute Zones backed by decarbonized and economically competitive nuclear power, hosting Gigafactories operated by European entities (OVHcloud, Scaleway, Mistral AI, IONOS), would produce compute with an F_sov close to 1 — the only compute genuinely beyond the reach of Cloud Sovereignty Mandates.

France's key decision in 2026–2027 is therefore no longer simply 'how many GPUs?' but 'how many GPUs under French jurisdiction?' These two metrics can diverge considerably if investment attraction policy remains indifferent to the legal nationality of operators.

5.12.3 The DARE/RISC-V Project: From Ambition to Strategic Necessity

Within Scenario D's framework and a fortiori under Cloud Sovereignty Mandates, the DARE project (Digital Autonomy with RISC-V in Europe, EuroHPC JU, 2025) changes status: it is no longer a long-term ambition but a strategic necessity.[29] As long as Europe depends exclusively on NVIDIA/AMD GPUs for its AI compute, US control over these architectures creates a residual vulnerability even in Gigafactories operated by European entities — a firmware update imposed by NVIDIA under the location verification program could theoretically degrade the performance of European clusters.

This risk is speculative but not unreasonable: it illustrates the depth of the architectural dependency. Genuine strategic autonomy in AI compute ultimately requires the capacity to design independent accelerators — a horizon that the DARE project places at 2030–2032, well beyond the 2028 tipping point.

5.13 Synthesis: The Fourth Tipping Point and the Conditions for a Managed Decoupling

The Great Decoupling is not inevitable. It represents a conditional systemic risk whose activation depends on US political choices and the speed of Europe's response. To the three tipping points identified in 5.7.2, a fourth is added.

Fourth tipping point: the implementation of location verification features in advanced GPUs. If BIS and the US Department of Commerce — in accordance with the July 2025 AI Action Plan — succeed in deploying remote attestation mechanisms in H100/B200/GB300 chips by end of 2026–2027, the technical substrate for the Cloud-Nationality Pivot will be in place. The question will no longer be whether Cloud Sovereignty Mandates are technically feasible, but solely whether they are politically decided. This tipping point should be monitored from 2026: the first BIS/NIST calls for tenders on remote attestation standards for AI chips will constitute the early signal.

For Europe and France, the managed decoupling strategy rests on three interdependent pillars: (i) accelerating the deployment of sovereign compute (EU Gigafactories under European jurisdiction) to increase F_sov before Mandate activation; (ii) securing Tier 1 status within the American alliance to maintain unrestricted access to advanced chips, by accepting coordination on export controls; (iii) investing in architectural alternatives (DARE/RISC-V, Huawei Ascend as a short-term alternative for non-sensitive workloads) to reduce dependency on US GPUs in the medium term.

The year 2028 constitutes the moment of truth not only for the four initial scenarios (as identified in 5.7.3) but for a more fundamental question: will Europe be able to maintain access to cutting-edge AI compute in a world where the legal nationality of compute takes precedence over its physical location? The decisions made in 2026–2027 — CADA, Gigafactories, nuclear energy, alignment on export controls — will determine whether the Great Decoupling is for Europe an existential threat or an opportunity to consolidate its strategic autonomy.

Notes

[1] Epoch AI (January 2026), “Trends in AI Hardware and Compute.” The doubling every 7 months of AI chip production combines 1.6×/year in quantity and 1.6×/year in performance per chip. Even a slowdown to 12 months would imply a quadrupling by 2030.

[2] IEA (April 2025), Energy and AI, Paris. The 800–950 TWh projections correspond to the IEA’s medium and high scenarios, with the low end of the range assuming a slowdown in AI adoption.

[3] United States Supreme Court (February 20, 2026), Learning Resources Inc. v. Trump and V.O.S. Selections v. United States, 6-3 decision: “IEEPA does not authorize the President to impose tariffs.” See Tax Foundation (2026), Tariff Tracker.

[4] White House (January 14, 2026), Proclamation 11002, section (2): “By July 1, 2026, the Secretary shall provide me with an update on the market for semiconductors that are used in United States data centers, so that the President may determine whether it is appropriate to modify the tariff.”

[5] Tax Foundation (February 2026), op. cit. The US-EU agreement (August 2025) caps semiconductor tariffs at 15% for the EU, but Proclamation 11002 explicitly provides for “broader tariffs” possible after Phase 1. The “aggressive” pole assumes a breach of this agreement.

[6] European Commission (2025), AI Continent Action Plan. Objective: triple the EU’s data center capacity in 5–7 years. 19 AI Factories selected, up to 5 Gigafactories (100,000+ GPUs each) planned. InvestAI fund: €20 billion to catalyze private investment. CFG (October 2025), “Special Compute Zones”: 180-day permitting, converted industrial zones.

[7] Tax Foundation (February 2026), op. cit. The US-EU agreement of August 2025 includes a 15% cap on semiconductor tariffs for the EU, reduction of auto tariffs from 27.5% to 15%, and negotiated sectoral exemptions.

[8] EuroHPC JU (2025). The 19 AI Factories plan up to 25,000 GPUs each (standard sites). Even at full capacity, this represents ~475,000 GPUs, less than the xAI Colossus cluster alone (200,000 H100 GPUs, expandable). Segler Consulting (June 2025) estimates total EU public capacity at ~57,000 accelerators in 2025.

[9] Proclamation 11002, section on the tariff offset program: “a tariff offset program to incentivize domestic manufacturing as previously announced.” Snell & Wilmer (February 2026), “The Continued Utilization of Tariffs to Control the Semiconductor Industry.”

[10] McKinsey Global Institute (May 2024), op. cit. The 0.3% figure corresponds to the “slow adoption” scenario, close to today’s level of productivity growth in Western Europe.

[11] Council of the EU (December 2025), adoption of the position on the amended regulation for AI Gigafactories. The provisional timeline places the first calls for tenders in late 2025 and the first operational installations in 2027–2028.

[12] CFG (October 2025), “Tripling the EU’s Data Centre Stock with Special AI Compute Zones.” The proposal advocates reuse of decommissioned industrial sites (former coal plants) with heavy grid connections, such as in Greece (former lignite mines converted).

[13] EuroHPC JU (March 2025), DARE project (Digital Autonomy with RISC-V in Europe), a 6-year program to develop integrated circuits based on the RISC-V processor. Three processor projects by distinct companies.

[14] Proclamation 11002, section (2): the Secretary of Commerce and the USTR must provide a report on the status of negotiations within 90 days, i.e., April 14, 2026. Pillsbury Law (January 2026), “Trump Admin Targets Advanced AI Semiconductors, Defers Broader Tariffs.”

[15] CFG (October 2025), op. cit. The average permitting time for a data center in the EU is currently 24+ months (versus 6–12 months in the United States). The SCZ proposal would reduce this to 180 days via a “single-window” process.

[16] RTE (2024), Futurs énergétiques 2050, scenario “N03.” France plans +10 GW of demand for data centers by 2030. The French nuclear advantage (65–70% of the mix) is unique in Europe and represents the main competitive differentiating factor for attracting AI investments.

17 Clarifying Lawful Overseas Use of Data (CLOUD) Act, Pub. L. 115-141 (March 23, 2018), Title III, § 103(a). Provision codified at 18 U.S.C. § 2713.
18 Microsoft France, Tribunal judiciaire de Paris (2024) — acknowledgement that the company cannot 'guarantee data sovereignty for European customers in the event of a legally founded injunction in the United States'; cited in The Register (December 22, 2025), 'Europe gets serious about cutting US digital umbilical cord.'
19 Michael Kratsios, OSTP Director, public statements at the APEC Digital and AI Ministerial Meeting, Seoul (August 2025), cited in TechResearchOnline (August 5, 2025), 'US AI Chip Tracking Plan.'
20 Bureau of Industry and Security, U.S. Department of Commerce, Framework for Artificial Intelligence Diffusion, Federal Register vol. 90, no. 10 (January 15, 2025), ECCN 3A090.a — compute caps of 50,000 H100-equivalents per Tier 2 country, 1,700 H100-equivalents per entity.
21 BIS, 'Suspension of the Affiliates Rule for One Year' (November 10, 2025). The rule, though suspended, maintains its control architecture and can be reactivated by administrative decision.
22 Financial Times (July 25, 2025), 'Nvidia AI chips worth $1bn smuggled to China after Trump export controls' — report on circumvention circuits via Malaysia, UAE, and Singapore in the first 6 months of 2025.
23 White House, Executive Order on Promoting the Export of the American AI Technology Stack (July 23, 2025). The AI Action Plan details the 'carrot and stick' strategy: preferential access for aligned allies, Foreign Direct Product Rule and secondary tariffs for non-cooperating countries.
24 US Department of Commerce, G42/Microsoft agreement (2024) — G42 (Abu Dhabi) agreed to divest from its partnerships with Chinese entities as a condition of access to advanced NVIDIA chips. Confirmed by statements from Commerce Secretary Gina Raimondo.
25 European Commission, Work Programme 2026, Cloud and AI Development Act (CADA), planned Q1 2026. See eu-cloud-ai-act.com for legislative tracking.
26 European Commission, Cloud Sovereignty Framework (October 2025), levels SOV-1 to SOV-3, published as part of the Cloud III Dynamic Purchasing System. Level SOV-3 requires full protection against non-European extraterritorial legislation.

27 Cristina Caffarra, Eurostack Foundation, cited in The Register (December 22, 2025): 'A company subject to the extraterritorial laws of the United States cannot be considered sovereign for Europe.'
28 See footnote 18.
29 EuroHPC JU, DARE project (Digital Autonomy with RISC-V in Europe), launched March 2025, 6-year program. Three AI accelerator projects based on the open RISC-V architecture, targeting first commercial availability around 2030–2031.

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